1.2 How Are Cells Adapted to Their Functions?

Biology · Cell Structure and Organisation · Syllabus refs (G2 & G3): 1d

Name: _____ Class: _____ Date: _____

Learning outcome

- Explain how the structures of specialised cells are adapted to their functions (e.g. muscle cell: many mitochondria to supply more energy; root hair cell: large surface area of cell membrane for greater absorption; red blood cell: lack of nucleus allowing it to transport more oxygen).

Why are cells specialised?

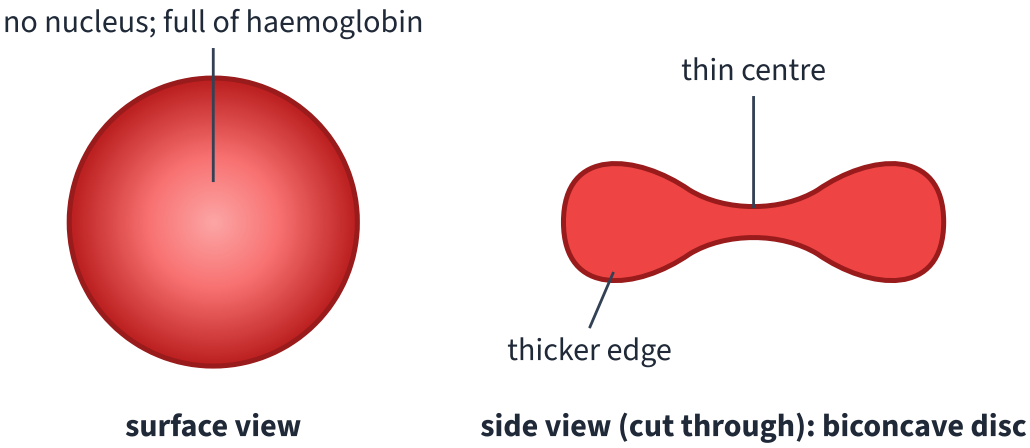
In a large organism, different cells do different jobs. A **specialised cell** has a structure that is **adapted** (suited) to its function. Each one still has the usual cell parts, but some parts are changed, added or lost.

> **Key idea:** for every adaptation, say **what the feature is** and **how it helps the cell do its job**.

Surface area to volume ratio

Substances such as oxygen, water and mineral salts enter a cell through its **surface** (the cell membrane). The more surface a cell has for its size, the faster it can take substances in. We compare cells using the **surface area to volume ratio** (SA : V).

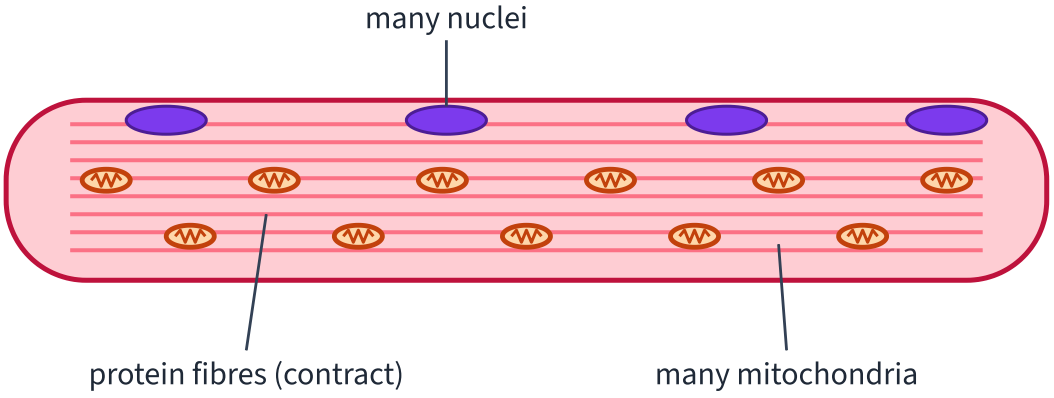
Red blood cell



A red blood cell. Function: **transports oxygen** around the body.

Adaptation	How it helps
Contains **haemoglobin**	Haemoglobin binds to oxygen, so the cell can carry oxygen around the body.
Circular, biconcave shape	Increases the surface area to volume ratio, so oxygen is taken in and released faster.
No nucleus	More space to pack in haemoglobin, so it can transport more oxygen.
Flexible	It can squeeze through narrow blood capillaries.

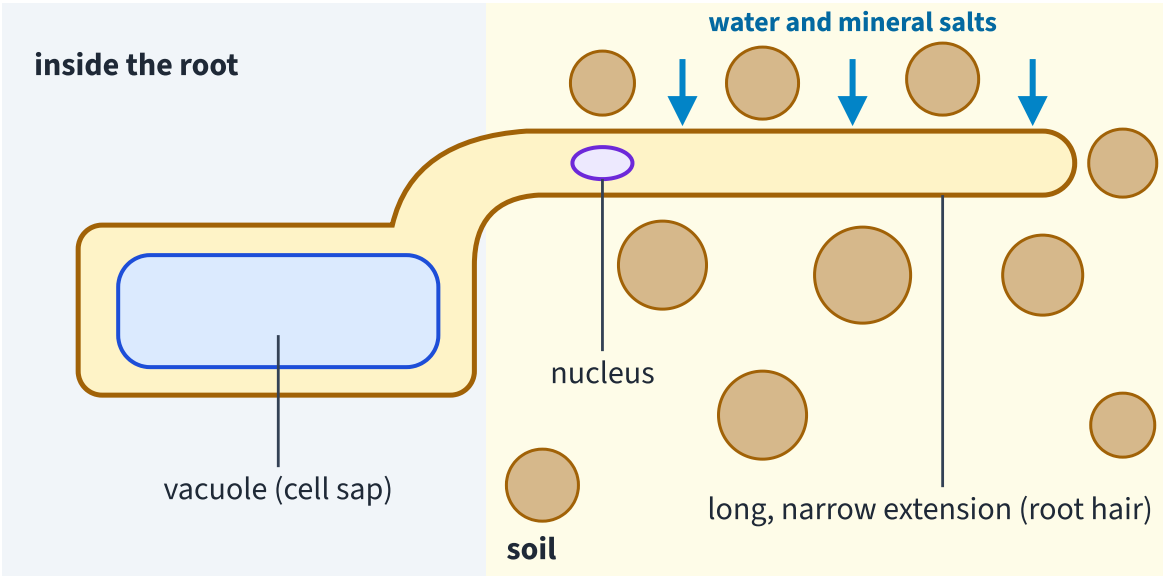
Muscle cell



A muscle cell. Function: **contracts and relaxes** to bring about movement.

Adaptation	How it helps
Contains **many mitochondria**	More aerobic respiration, so more energy is released for contraction.
Has **contractile protein fibres**	The fibres contract and relax to bring about movement.
Has **many nuclei**	Allows for cell division, growth and repair of the long cell.

Root hair cell



A root hair cell. Function: **absorbs water and mineral salts** from the soil.

Adaptation	How it helps
Long and narrow extension (the root hair)	Increases the surface area to volume ratio, so water and mineral salts are absorbed faster.
Cell sap in the vacuole has a **lower water potential** than the soil water	Water enters the root hair cell by **osmosis**. (You will learn about osmosis in Topic 2.)
No chloroplasts	It grows underground with no light, so chloroplasts would have no use.

Writing about adaptations

Link the **feature** to its **effect** and the **function**:

feature → so that… → which helps the cell to…

Example: “A red blood cell has **no nucleus**, **so** there is more space for haemoglobin, **so** it can transport more oxygen.”

Check your understanding

1. Why does a red blood cell have no nucleus?

- ○ So it can divide faster
- ○ To make room for more haemoglobin
- ○ So it can absorb water
- ○ To release more energy

2. Muscle cells contain many mitochondria. What is the reason?

- ○ To make more proteins
- ○ To store cell sap
- ○ To release more energy by aerobic respiration for contraction
- ○ To absorb more light

3. How does the long extension of a root hair cell help the plant?

- ○ It gives the cell a fixed shape.
- ○ It increases the surface area to volume ratio for absorbing water and mineral salts.
- ○ It carries oxygen to the leaves.
- ○ It makes food by photosynthesis.

4. A cube-shaped cell has sides of 2 µm. Its surface area is 24 µm^2 and its volume is 8 µm^3. Work out the surface area to volume ratio (as a number : 1).

Your answer (: 1)

5. The biconcave shape of a red blood cell…

- ○ lets it squeeze through capillaries.
- ○ increases its surface area to volume ratio for faster uptake of oxygen.
- ○ gives it room for a nucleus.
- ○ makes it contract.